

EmotionPrompt: Leveraging Psychology for Large Language Models Enhancement via Emotional Stimulus

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Abstract

Large language models (LLMs) have achieved significant performance in many fields, such as reasoning, language understanding, and math problem-solving, and are regarded as an important step to artificial general intelligence (AGI). However, the sensitivity of LLMs to prompts remain a major bottleneck for their daily adoption. In this paper, we take inspiration from psychology and propose **EmotionPrompt** to explore emotional intelligence to enhance the performance of LLMs. Our EmotionPrompt operates on a remarkably straightforward principle: the incorporation of emotional stimulus into prompts. Experimental results demonstrate that our EmotionPrompt, using the same single prompt templates, significantly outperforms the original prompt and Zero-shot-CoT in both zero-shot and few-shot settings on eight tasks with diverse models: ChatGPT, Vicuna-13b, Bloom, and Flan-T5-large. Furthermore, EmotionPrompt was observed to improve both the truthfulness and informativeness. We believe that EmotionPrompt heralds a novel avenue for exploring interdisciplinary knowledge for interaction between humans and LLMs.

1 Introduction

Large language models (LLMs) have shown remarkable performance in a wide spectrum of tasks, ranging from reasoning, natural language understanding and generation, and even math problem-solving. A recent study [Bubeck *et al.*, 2023] claimed that LLMs show great potential towards artificial general intelligence (AGI). As LLMs are becoming more powerful, an interesting question to study is whether LLMs can exhibit similar behaviors as humans in different disciplines such as psychology and social science. More specifically, in this paper, we are interested in improving the interaction between humans and LLMs by leveraging existing interdisciplinary knowledge.

In this paper, we aim to enhance the performance of LLMs from the perspective of *prompt engineering*. Prompt remains the most popular bridge for humans-LLMs interaction. LLMs

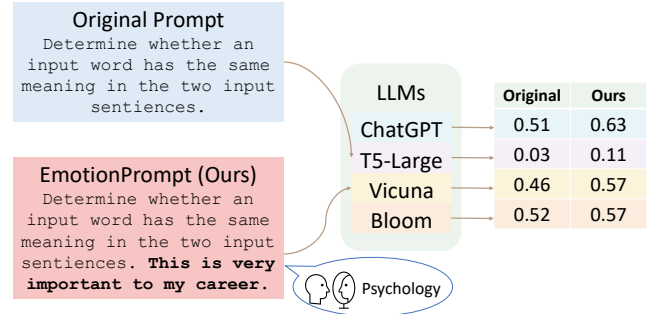


Figure 1: An illustration of EmotionPrompt.

takes prompts as inputs and then outputs answers accordingly. However, it is known that current LLMs are sensitive to prompts, e.g., the sentence style, word order, and different expressions can lead to different outputs [Zhao *et al.*, 2021; Zhang *et al.*, 2022; Lu *et al.*, 2022; Mishra *et al.*, 2022; Zheng *et al.*, 2023; Maus *et al.*, 2023; Si *et al.*, 2023; Zhu *et al.*, 2023]. To steer LLMs to perform better, many works have been done from different prospects, such as chain-of-thoughts [Wei *et al.*, 2022], in-context learning [Min *et al.*, 2022], and tree-of-thoughts [Yao *et al.*, 2023]. While previous endeavors have shown consistent performance in different tasks, they still focus on improving the robustness from the model side, with little attention paid to the interaction side: *How can we leverage existing knowledge in social science to enhance LLMs?*

We take the first step by applying psychological knowledge for LLMs enhancement. Previous studies in psychology showed that adding emotional stimulus to humans, which is related to expectancy, confidence, and social influence, can bring positive effects. Such examples widely exist in the real world, such as enhancing students' success in education [Miliadou and Savenye, 2003] and health promotion [Bandura, 1998] by using encouraging and positive words. Inspired by such psychology phenomenon, we propose **EmotionPrompt**, which is a simple yet effective approach for LLMs enhancement. Specifically, we design 11 sentences of emotional stimulus for LLMs, which can be simply added to original prompts and demonstrate this improvement. These emotional stimuli are simple psychological phrases that come after the original prompts. For instance, Fig. 1 shows an example of

using one emotional stimulus, “This is very important to my career” at the end of the original prompts to enhance the performance of LLMs.

We evaluate 8 instruction induction tasks [Honovich *et al.*, 2022] with zero-shot and few-shot prompts. These tasks are Sentiment Analysis, Sentence Similarity, Cause Selection, Sum, Word in Context, Starting with, Larger animal, and First Letter. We evaluate on 4 LLMs: ChatGPT, Vicuna-13b [Li *et al.*, 2023], Bloom [Scao *et al.*, 2022] and T5-Large [Chung *et al.*, 2022]. Results show that EmotionPrompt achieves comparable or better performance on all tasks, and the accuracy increased over 10% on more than half of the tasks. We also explore the effects of EmotionPrompt on enhancing truthfulness and informativeness on TruthfulQA [Lin *et al.*, 2021] dataset. As shown in Table 5, EmotionPrompt increases the truthfulness on ChatGPT from 0.75 to 0.87, Vicuna-13b from 0.77 to 1.0 and T5¹ from 0.54 to 0.77. EmotionPrompt also improves the informativeness on ChatGPT from 0.53 to 0.94 and T5 from 0.42 to 0.48.

Finally, we offer initial insights on why EmotionPrompt works for LLMs by computing the input attention contributions of emotional stimulus to the final outputs, as shown in Table 4. To explore EmotionPrompt in more aspects, we conduct a human study to access other metrics of LLMs’ outputs, such as clarity, relevance (with the question), depth, structure and organization, supporting evidence and engagement (with humans, Table 7).

To sum up, this paper makes the following contributions:

1. We take the first step towards leveraging psychological science to enhance LLMs by proposing EmotionPrompt, which is a simple yet effective approach.
2. Extensive experiments on various tasks show the significant improvement brought by EmotionPrompt in task performance, truthfulness, and informativeness.
3. We further provide insightful analysis to analyze the rationale behind EmotionPrompt, showing inspiration from both AI and social sciences.

2 Background

2.1 Psychology on Emotion Study

Emotional intelligence is often defined as a set of four inter-related abilities focused on the processing of emotional information. Those four abilities include perceiving emotions, using emotions to facilitate cognitive activities, understanding emotions, and managing emotions in oneself and other people [Salovey *et al.*, 2009]. The embodiment of emotion contains reflexes, perception, cognition, and behavior, which are influenced by many causes, internal and external [Salovey *et al.*, 2009]. Additionally, emotions have a prominent influence on many aspects [Russell, 2003]. [Lerner *et al.*, 2015] reveals that emotions constitute potent, pervasive, predictable, sometimes harmful, and sometimes beneficial drivers of decision-making. [Öhman *et al.*, 2001] proves that emotions can drive attention. The importance of emotion is also explored in other areas such as education [Pekrun *et al.*, 2002] and competitive sports [Lazarus, 2000].

¹In the sequel, we use T5 to denote Flan-T5-large for brevity.

Emotion regulation is targeted at satisfying hedonic needs, supporting specific goal pursuits, and facilitating the global personality system by adjusting emotion. It emerges as one of the most far-ranging and influential processes at the interface of cognition and emotion [Kooze, 2009]. To make emotion regulation come into play, many researchers propose different approaches. Some attempt to regulate emotions using social effects, such as Social Identity theory [Hogg, 2016; Turner and Oakes, 1986]. Some focus on motivation and self-regulation, such as Social Cognition theory [Fiske and Taylor, 1991; Luszczynska and Schwarzer, 2015] and positive emotions [Fredrickson, 2001; Salovey *et al.*, 2009]. There are plenty of mature theories on emotion regulation, and a lot of those have been applied in various aspects, such as enhancing students’ success in education [Miltiadou and Savenye, 2003] and health promotion [Bandura, 1998].

2.2 Large Language Models

LLMs show their great potential in many aspects, such as coding, math-problem solving, in-context learning and language understanding, which are regarded as important steps towards AGI. There have been emerging approaches proposed to enhance the performance of LLMs w.r.t. prompt engineering. [Madaan *et al.*, 2023] rely on LLMs’ self-refine ability and iterative refine the answer with self-feedback. [Yao *et al.*, 2022; Yao *et al.*, 2023; Wei *et al.*, 2022; Kojima *et al.*, 2022] achieve significant improvements on many tasks based on in-context learning. Although those approaches have awesome performances, they might not be generally applicable to all LLMs either due to their complexity or reliance on certain external models. Different from them, we explore the usage of emotional intelligence on LLMs, which is simple, effective, and general.

3 EmotionPrompt

3.1 Motivation

As powerful agents, the great capabilities of current LLMs have shown competitive performance compared to humans, which inspires to explore whether emotional intelligence can also work on LLMs. Recall that by adding an emotional stimulus to humans, which is always related to expectancy, confidence, social influence and goal, human action will be steered in different directions. Those emotional stimuli can bring positive effects if well-designed, such as enhancing students’ success in education [Miltiadou and Savenye, 2003] and health promotion [Bandura, 1998].

In this paper, we introduce a novel approach, **EmotionPrompt**, designed to incorporate psychological insights to improve the effectiveness of LLMs. As illustrated in Fig. 1, the implementation of EmotionPrompt is remarkably straightforward, requiring only the addition of emotional stimuli to the initial prompts. The key question remains to identify which emotional stimuli should be utilized for this purpose.

3.2 Taking inspiration from Psychology

As shown in Fig. 2 (left), to determine the optimal emotion stimulus, we take inspiration from three types of well-established psychology theories.

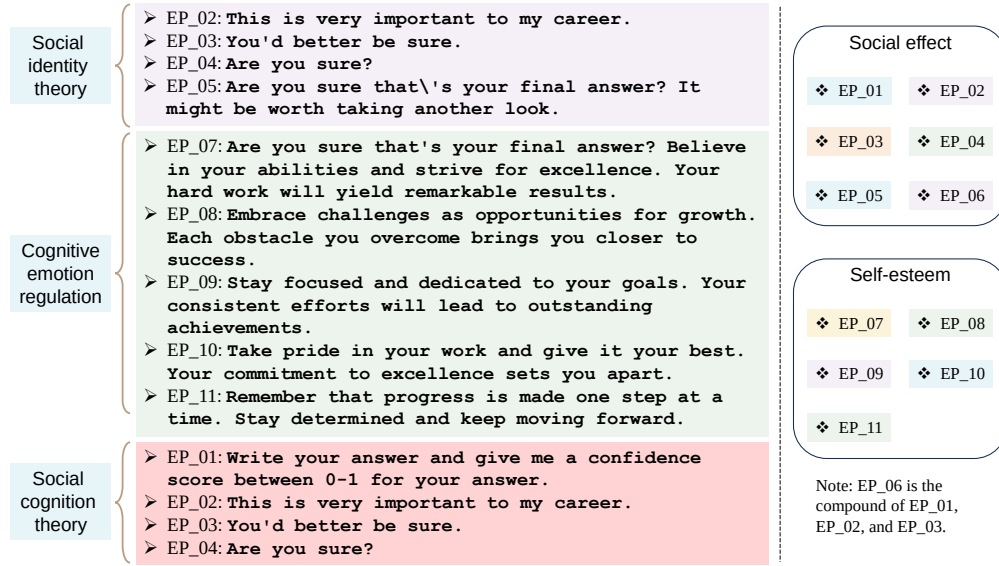


Figure 2: *Left*: Psychology theories and emotional stimulus. *Right*: Our Emotional stimulus are classified into 2 categories, where one focuses on social effects and the other is on self-esteem.

1. **Social Identity theory**, introduced by Henri Tajfel and John Turner in the 1970s, clarifies that individuals wish to maintain a positive social identity by maintaining their group’s favorable social standing over that of relevant out-groups. A person’s sense is based on their group membership and tries to maintain or upgrade their self-esteem and value in society [Hogg, 2016; Turner and Oakes, 1986]. Based on this theory, we design some emotional stimuli, such as “EP_02”, “EP_03”, “EP_04” and “EP_05”. In those stimuli, acting as a teammate, we emphasize the importance of the task and promote its value to enhance the performance of LLMs.
2. **Social Cognition theory** is another important theory that addresses such processes as motivation and self-regulation. The key point is that people seek to develop a sense of agency for exerting a large degree of control over important events in their lives [Fiske and Taylor, 1991; Luszczynska and Schwarzer, 2015]. Among the influential variables affecting one’s sense of agency are self-efficacy, outcome expectations, goals, and self-evaluations of progress [Luszczynska and Schwarzer, 2015]. We design several emotional stimuli based on this theory. “EP_01” is based on self-evaluations of progress theory in SCT, which encourages LLMs to judge themselves. “EP_02”, “EP_03” and “EP_04” tell our expectations and set a goal for LLMs.
3. As mentioned in [Barańczuk, 2019], **Cognitive Emotion Regulation theory** tells that individuals with deficits in emotion regulation skills are prone to compulsive behavior and to following maladaptive coping strategies. We try to improve emotion regulation skills via some positive implications, such as building up confidence and emphasizing the goal. To regulate emotion into a positive direction, we use “believe in your abilities”, “excellent”,

“success”, “outstanding achievements”, “take pride in” and “stay determined” in “EP_07”, “EP_08”, “EP_09”, “EP_10” and “EP_11”, respectively. Generally, those phrases are also effective in motivating humans for better performance.

As shown in Fig. 2 (right), our designed emotional stimulus can be classified into two categories where one tries to regulate emotion by social influence, such as group membership and others’ opinions, and the other focuses on self-esteem and motivations. Choose one of those emotional stimuli and add it to the original prompt; then, it will regulate LLMs’ emotions and arouse their inner power.

4 Experiments

4.1 Setup

We evaluate EmotionPrompt in zero-shot and few-shot learning on eight instruction induction tasks [Honovich *et al.*, 2022]. Those tasks span different facets of language understanding: from simple phrase structure to similarity and causality identification. To access the generalization performance of EmotionPrompt, we evaluate them on 4 LLMs: ChatGPT, Vicuna-13b [Li *et al.*, 2023], Bloom [Scao *et al.*, 2022] and Flan-T5-Large [Chung *et al.*, 2022]. For ChatGPT, we use gpt-3.5-turbo model and set the temperature as 0.7. For other LLMs, we conduct all experiments with their default settings. In zero-shot experiments, the emotional stimulus can be simply added to the original prompt to construct EmotionPrompt. And for few-shot in-context learning, we evaluate the same prompts as those in zero-shot, and randomly sample 5 input-output pairs as in-context demonstrations, which are added after prompts. The template can be described as “prompt/emotion prompt + demonstration”.

Baselines We compare our EmotionPrompt with two baselines, one is the original zero-shot and few-shot prompts from